

EMCal calibration update: data/sim comparisons

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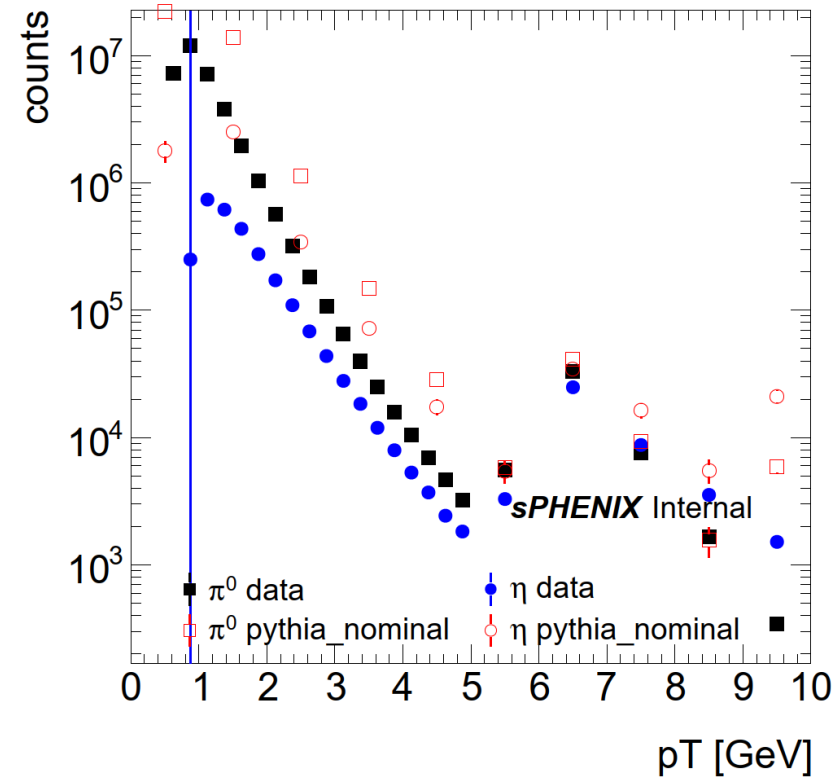
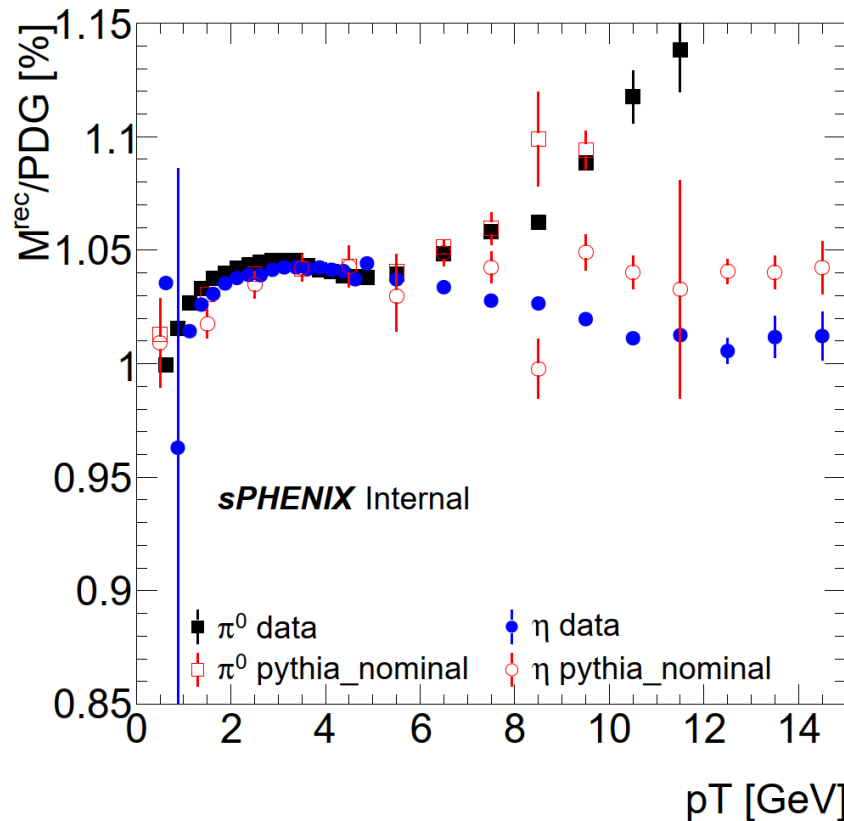
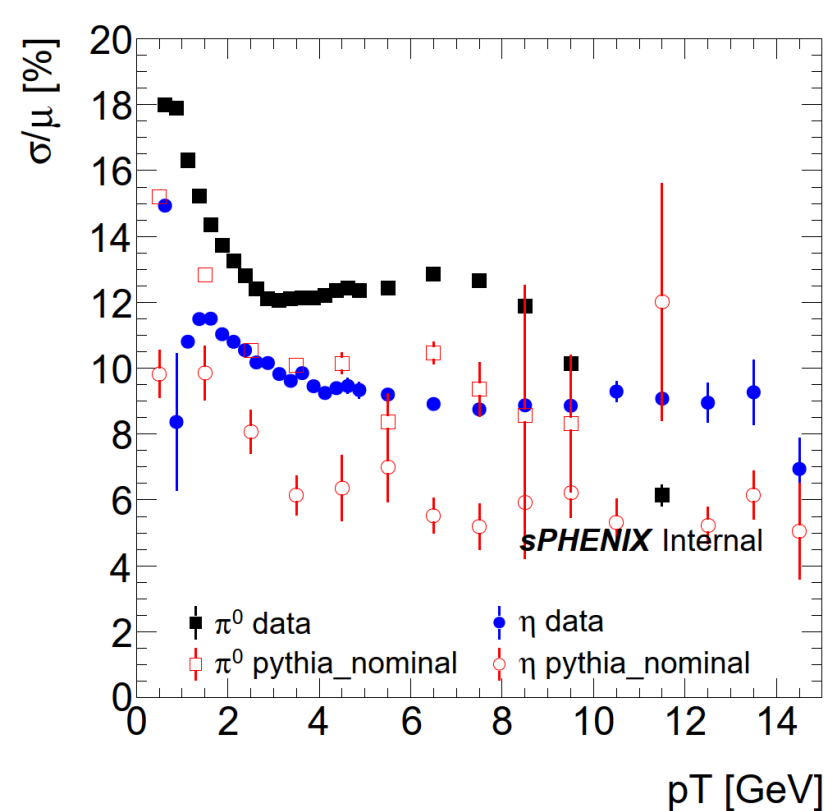
Selections

2

- **Data: ana509, GRL**
- **Pythia run 28: main samples MB pythia, Eta3 and Eta8**
 - Other samples JetXX, Jet12+MB (double interactions)
- **MBD vertex $|z| < 20$ cm**
- **Cluster selections**
 - $p_{T1,2} > 0.25$ GeV
 - $Asym < 0.6$
- **New since last week**
 - New high p_T eta samples

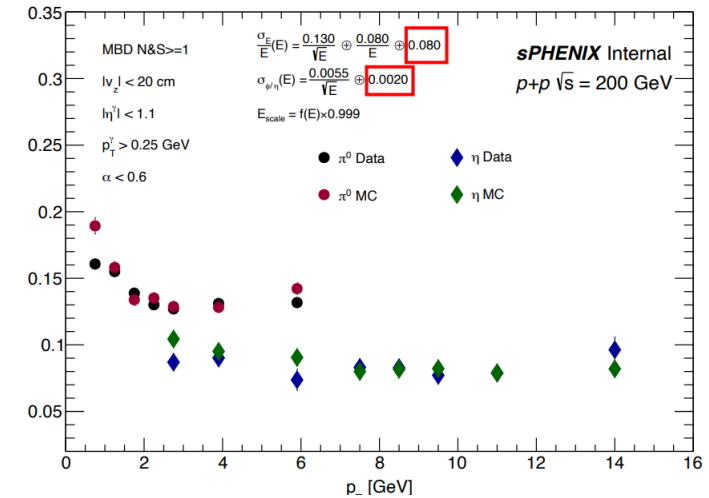
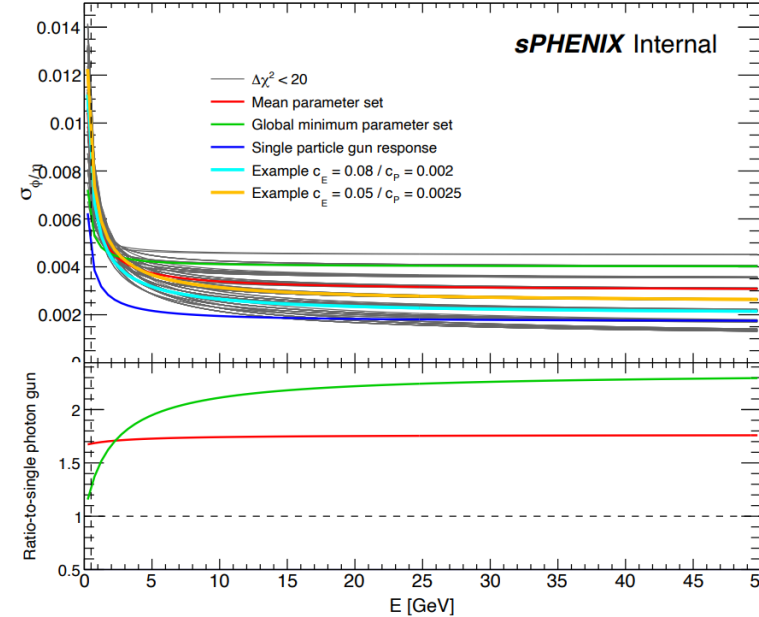
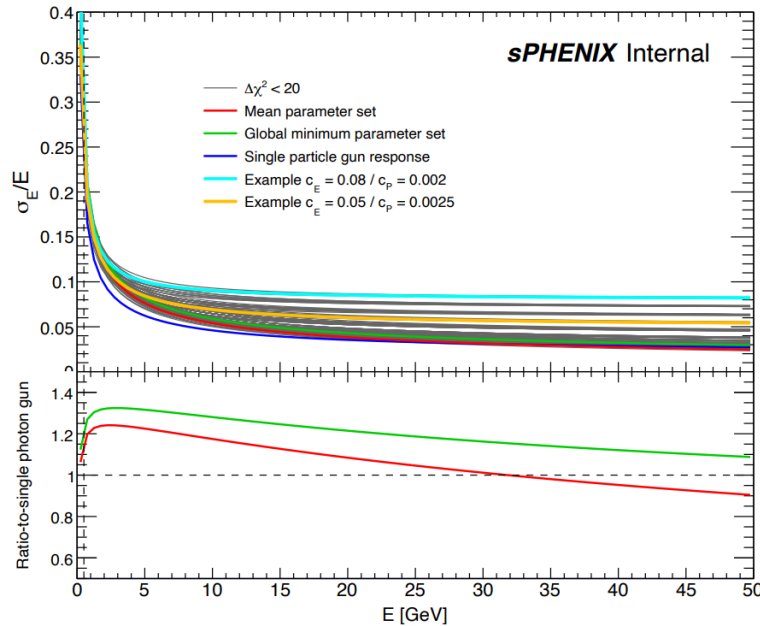
Data pythia comparison

- Pythia: MB + jet5 ($p_T > 6$ GeV) + jet8 ($p_T > 9$ GeV) + jet12 ($p_T > 12$ GeV)
- Data is much wider than sim...
- It is unclear why, and thus we are attempting to smear the MC

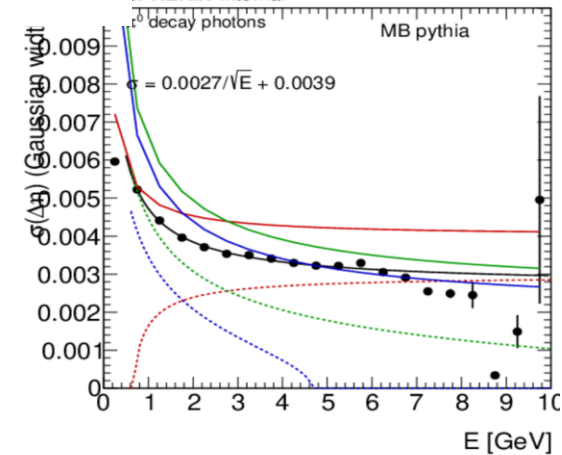
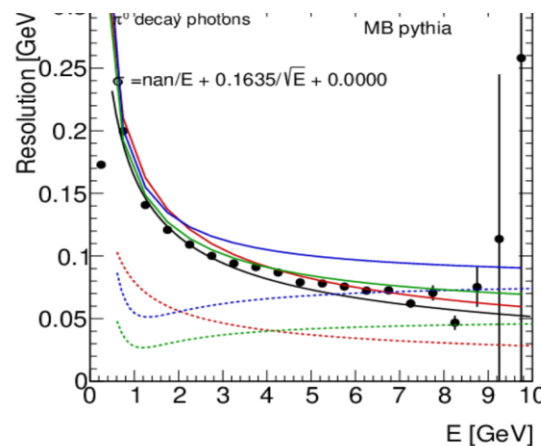


Data-driven extraction of resolution

➤ https://indico.bnl.gov/event/31859/contributions/120756/attachments/68548/117758/EMCal_toyMC_calib_20260302.pdf

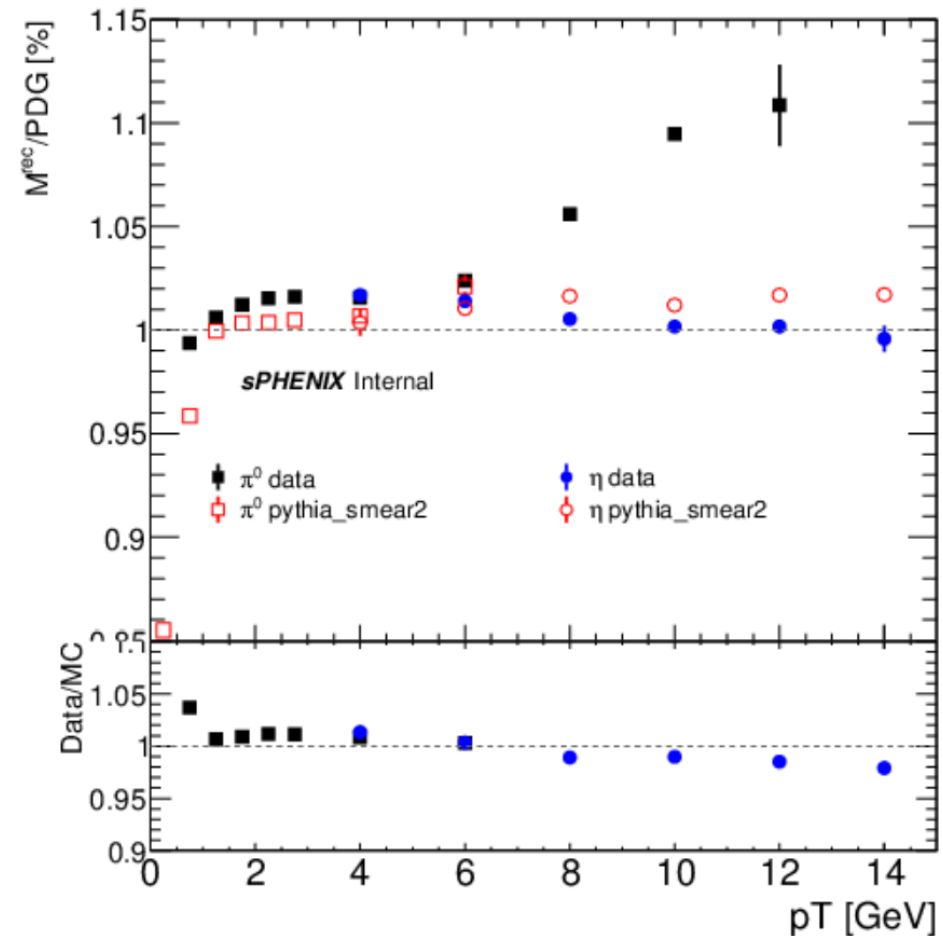
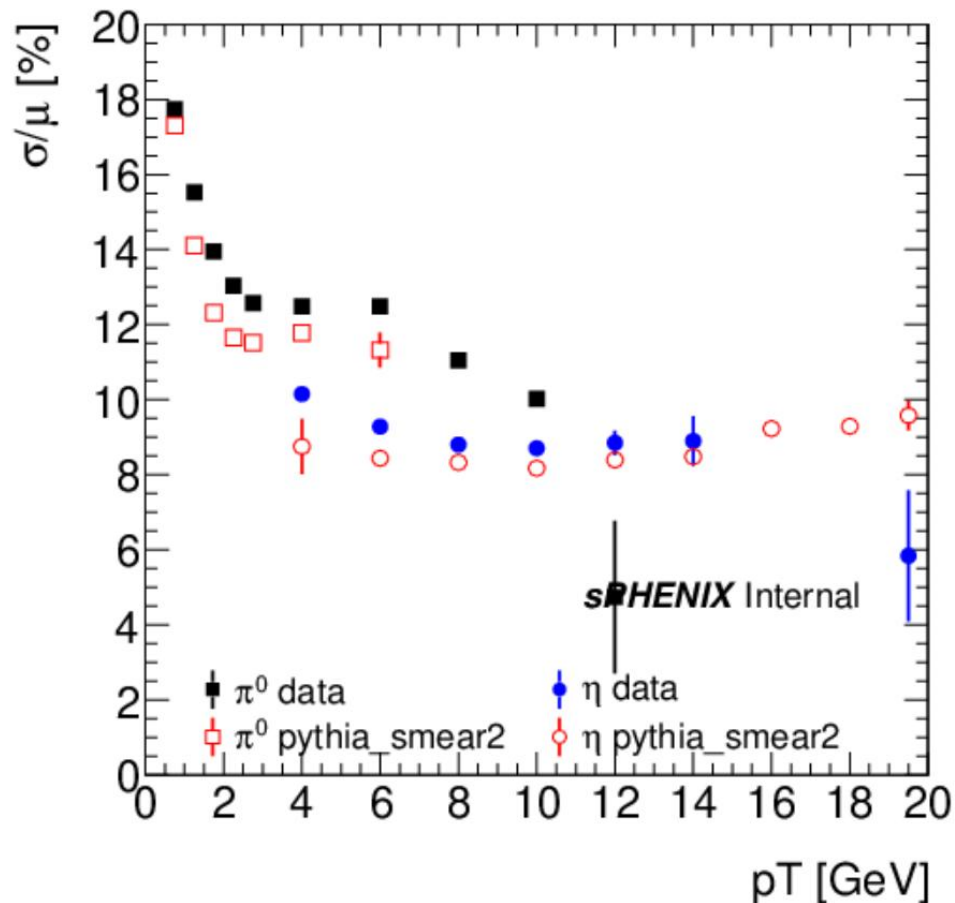


11 smearing is needed to reproduce the data



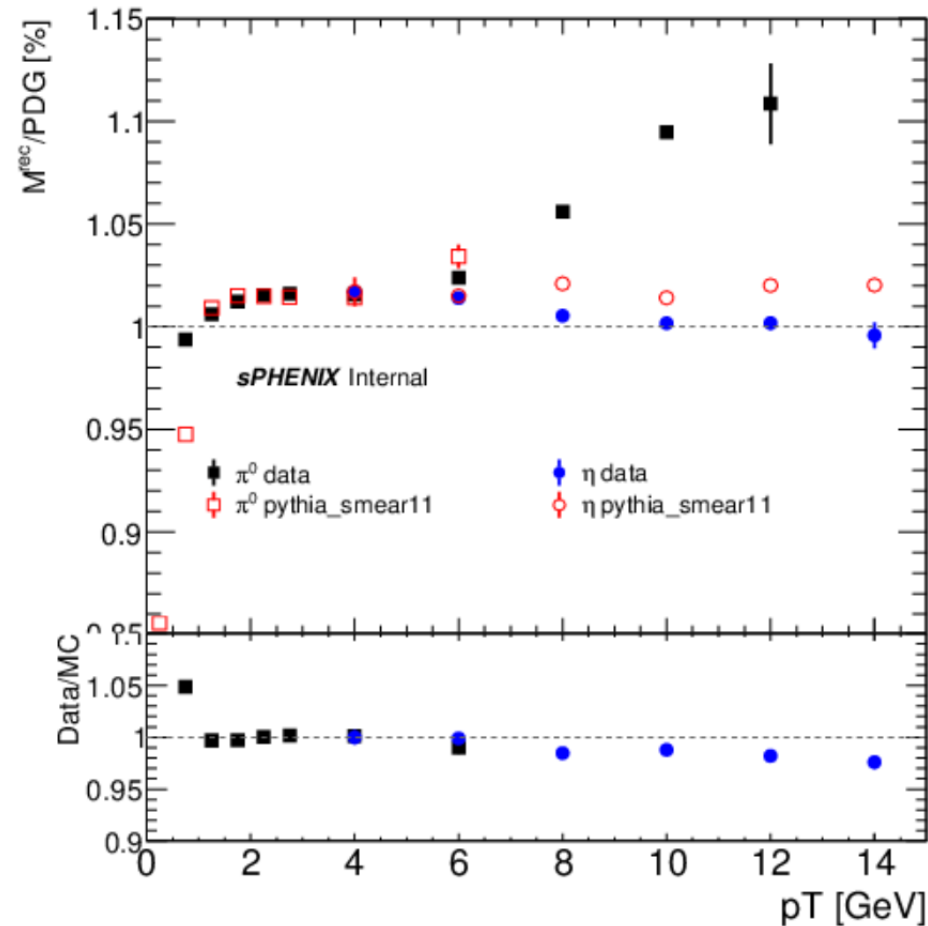
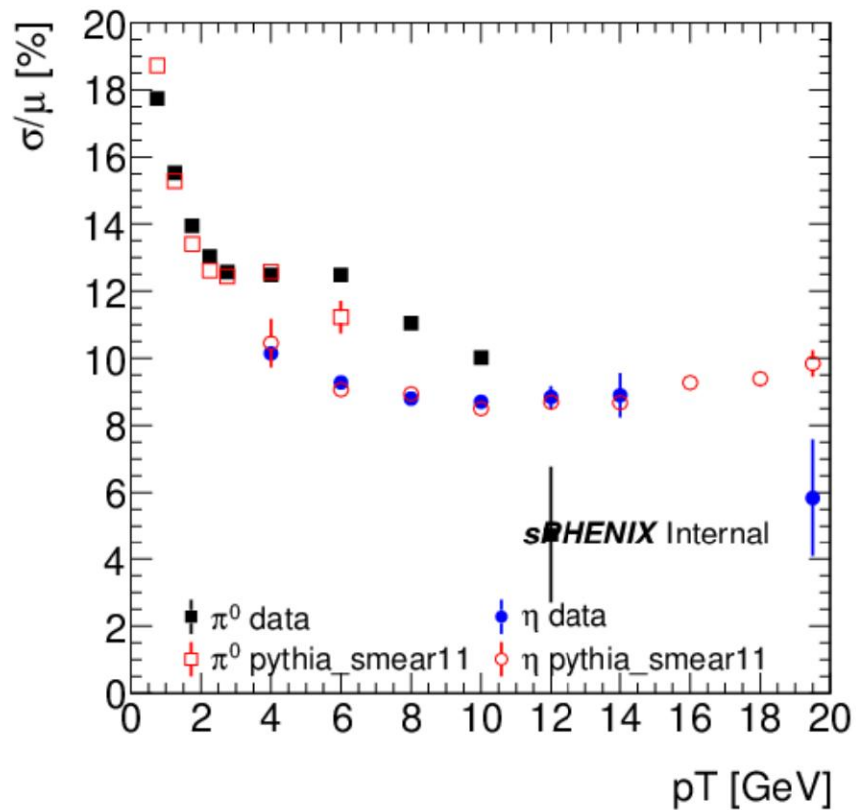
Post smearing (first attempt) data/MC

- This well defined process did not seem to work perfectly and the smearing factors was insufficient to reproduce the data



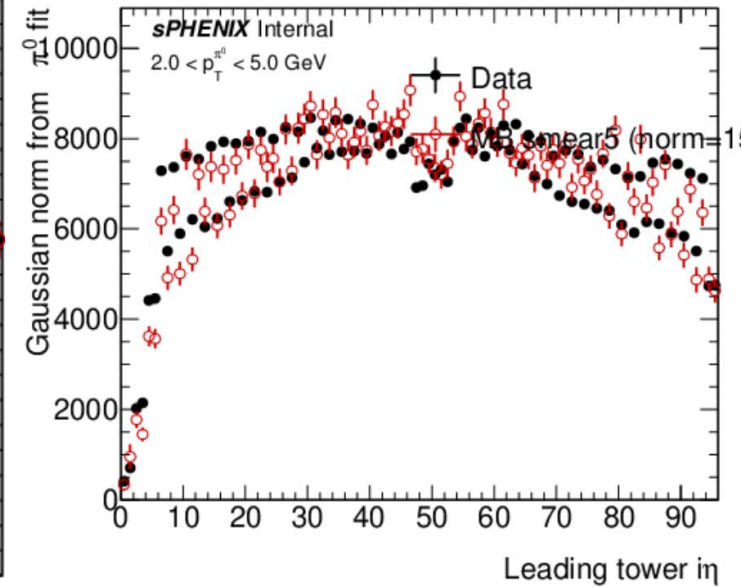
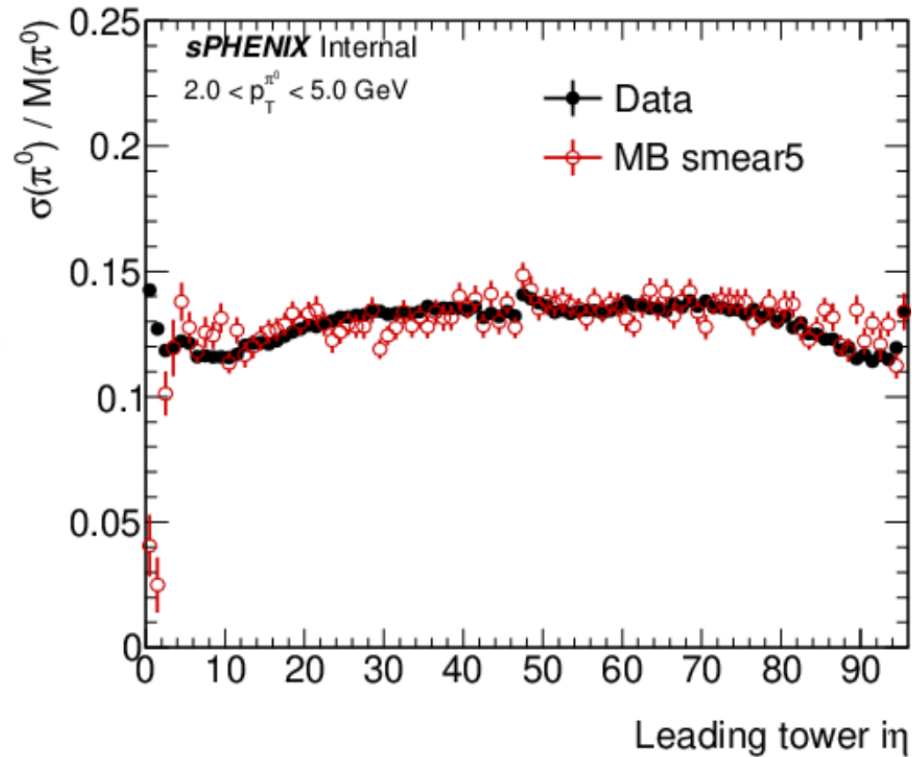
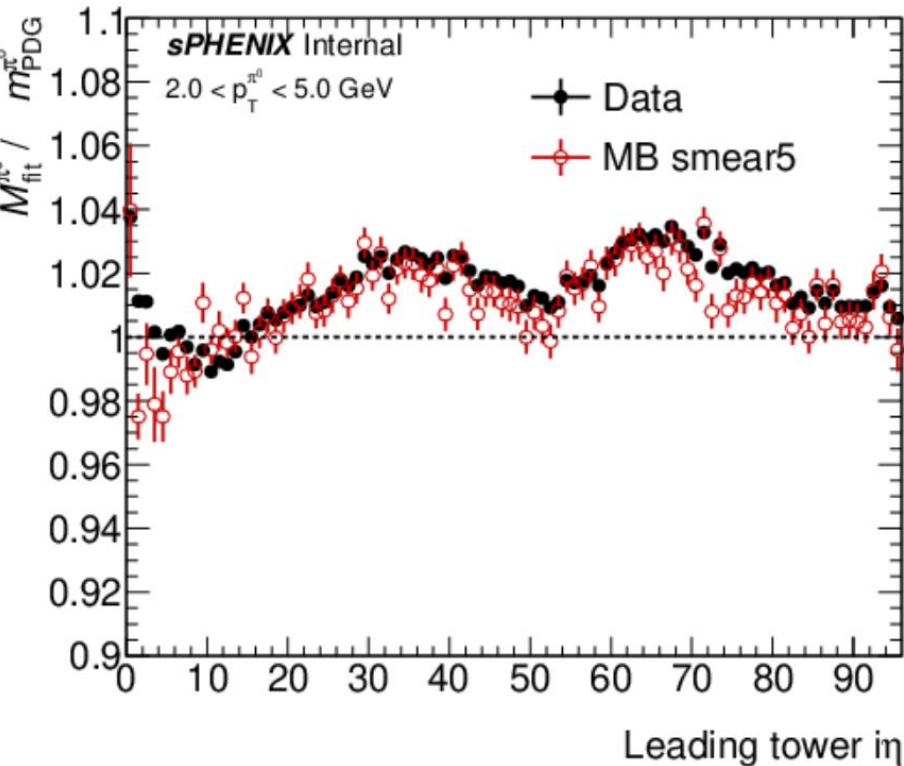
Post smearing (second attempt) data/MC

- Added an additional 8%/sqrt(E) of energy smearing
- 2% discrepancy at high pT in the eta



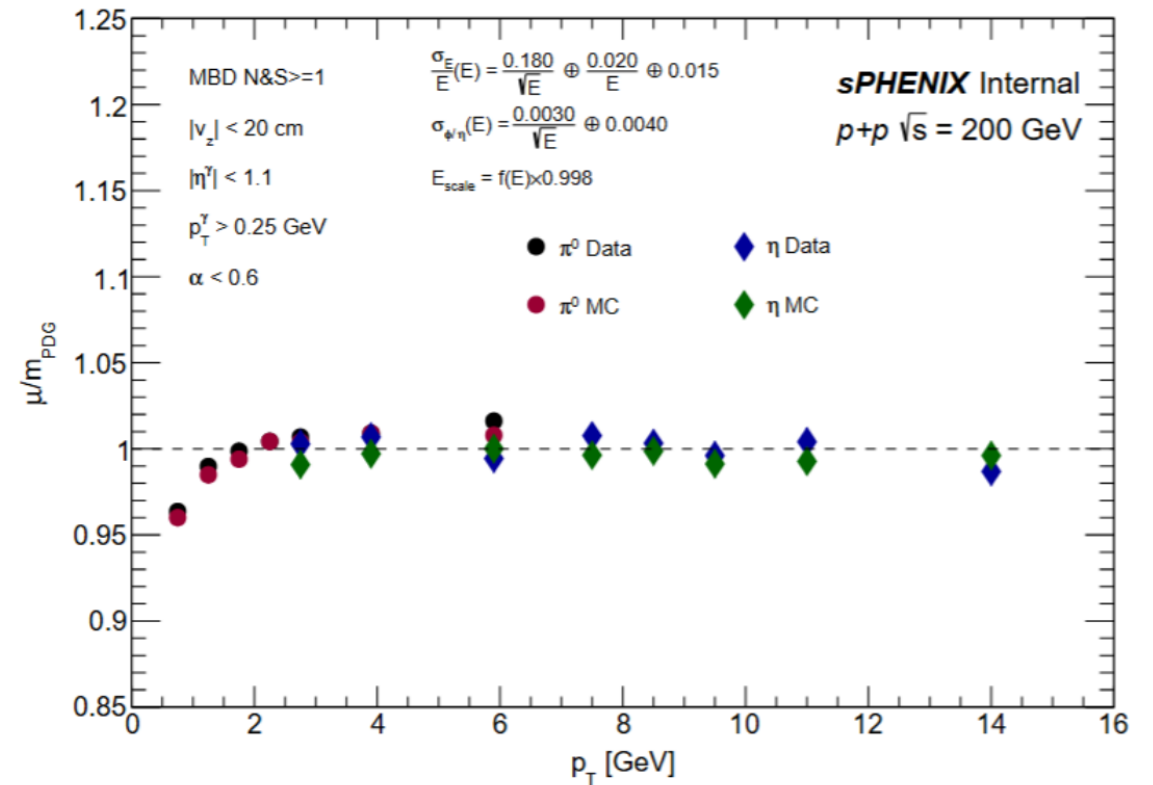
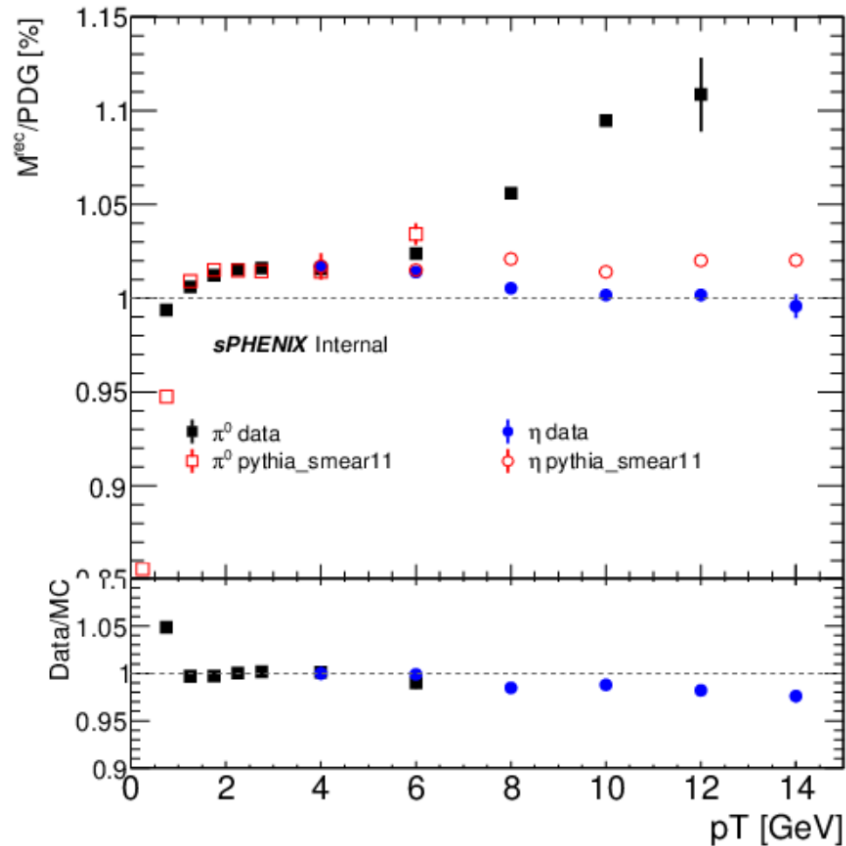
Calibration

➤ Set scale eta-by-eta



High pt eta meson

- Right: Jaebeom's instance of the dame plot
- Need more stats in MC

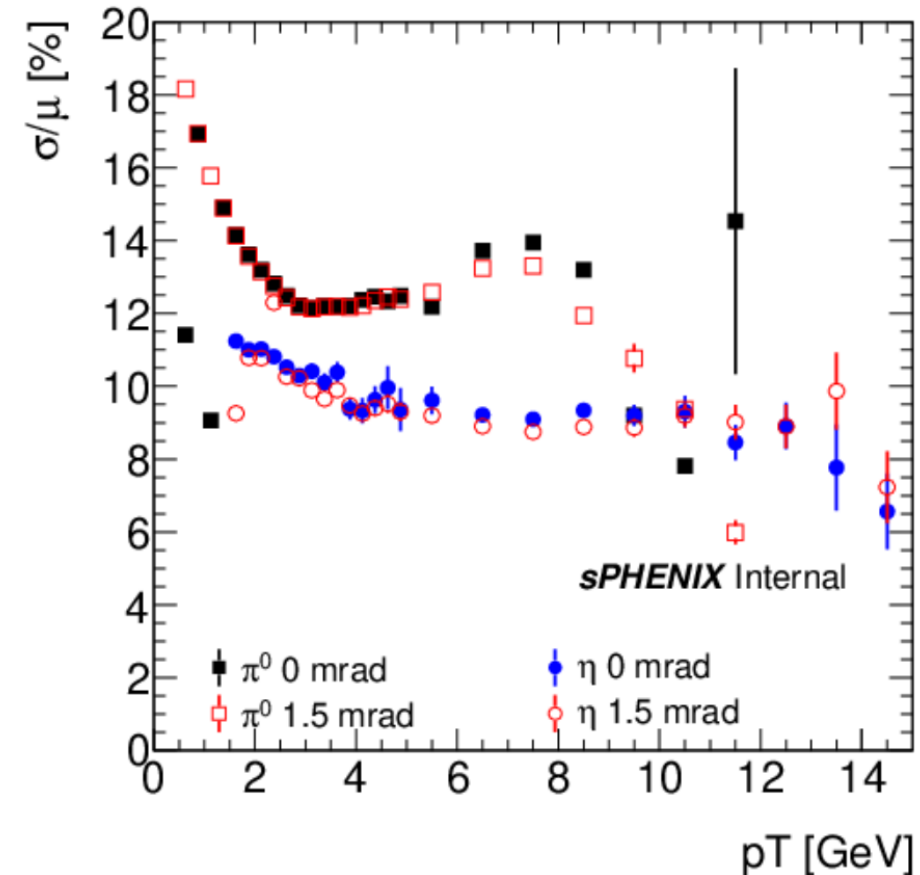
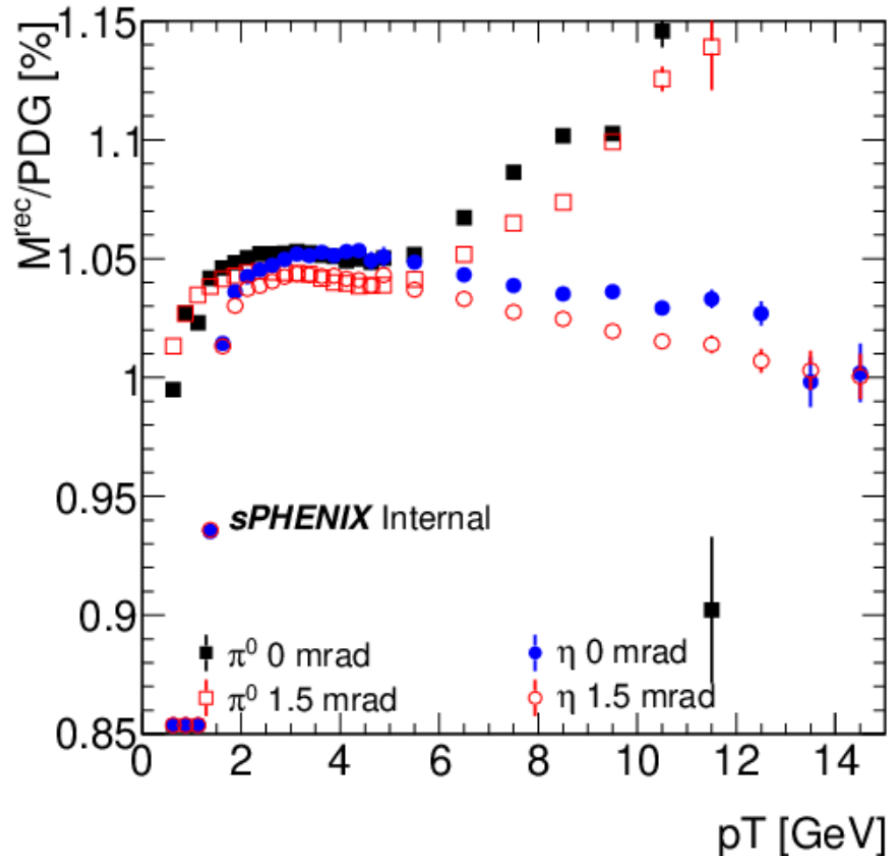


Conclusion

- Investigate apparent 2-2.5%% non-linearity
- Calibration set last week still good considering slightly different smearing scheme

0mrad vs. 1.5 mrad data

- 0mrad (1.5mrad) is estimated to have ~22% (7%) double interactions, which degrades vertex resolution
- This degradation is worse in 0mrad because the vertex width is ~50 cm where it is ~15cm in 1.5mrad
- For the rest of this presentation use 1.5 mrad for data/MC comparisons



Pileup effects in MC

- Large effect seen in double interaction sim
- Could be effect of having the wrong vertex or could be poor performance at large truth z vertex which enters the because of a bad reco that is less than 20

